COMPRESSION NUMBER THEORY ON THE FINITE PROVABILITY OF INFINITE NUMBER-THEORETIC PROPOSITIONS

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Abstract. This paper proposes an axiomatic meta-mathematical framework concerning the **finite provability** of infinite number-theoretic propositions, termed Compression Number Theory (CNT). The framework rests on a dichotomy of infinite sequences of natural numbers according to their information-compression property, thereby partitioning number-theoretic propositions into compressible and incompressible classes according to the structure of the sequences on which they rest. In this way it establishes a causal link from sequence compressibility to the finite provability of propositions. Within this framework we formalize Goldbach's conjecture, introduce the classification of *plausible-but-unprovable propositions*, and propose the **Ultimate Meta-Conjecture of CNT**, which captures the essential relation between the finite provability of Goldbach's conjecture and the postulates of the framework. CNT does not decide the truth values of number-theoretic propositions; it studies only whether they admit a finite mathematical proof.

Keywords: Compression Number Theory; finite provability; infinite sequence dichotomy; cross-domain matching proposition; plausible-but-unprovable proposition; Goldbach's conjecture
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1. Introduction

CNT does not decide the truth values of number-theoretic propositions; it studies only their finite provability. Whether a proposition holds and whether it admits a finite mathematical proof are questions of a different nature. The sole object of study of CNT is the latter, namely whether an infinite number-theoretic proposition admits a finite mathematical proof.

The central thesis of the framework is that the provability of a number-theoretic proposition is not structureless but is determined by the information-compression property of the infinite sequence on which it depends. If the truth of a proposition depends on an infinite sequence that cannot be generated as a whole by a static finite expression, then any proof of that proposition must process the infinitely many objects of the sequence point by point, and therefore can hardly be compressed into finite reasoning. CNT formalizes this intuition into a deductive system of postulates and theorems.

2. Notation and Preliminaries

- $\Box_{\text{fin}} P$: the proposition P admits a finite proof.
- Compressible(S): the sequence S is a CNT-compressible sequence.
- Incompressible(S): the sequence S is a CNT-incompressible sequence.
- CrossMatching(P): the proposition P is a cross-domain matching proposition.
- Prov(P): the proposition P is provable.
- $\mathbb{P}$: the set of prime numbers (a set).
- $S_{\mathbb{P}}$: the prime sequence, i.e. the increasing infinite sequence of primes $(2, 3, 5, 7, 11, \dots)$.

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3. Basic Definitions

3.1. **Finite Proof.**

Definition 3.1 (Finite proof)**.** A finite proof $\Box_{\text{fin}}P$ of a proposition P is a proof satisfying the following four conditions: (i) it uses finitely many symbols; (ii) it consists of finitely many inference steps; (iii) each step is mechanically verifiable; (iv) it does not depend on infinite enumeration.

Finiteness is the benchmark of the entire argumentation. Whenever a proposition is said to be provable, it means that there exists a finite proof satisfying the above conditions; whenever it is said to be unprovable, it means that no such finite proof exists.

3.2. **CNT Static Finite Generation Structure.**

Definition 3.2 (CNT static finite generation structure)**.** A structure consisting solely of purely static mathematical expressions that are free of branching, loops, enumeration, and primality testing is called a CNT static finite generation structure. Algorithms, sieves, iterative procedures, and conditional judgements do not belong to CNT generation structures.

By "static" we mean that an expression yields a definite output at each input position without relying on step-by-step execution, conditional transfer, or search processes. Dynamic processes such as sieves and iterations may in principle generate sequences, but they purchase covering power at the cost of running steps whose length grows uncontrollably with the input, and hence they are not static generation.

3.3. **CNT-Compressible Sequence.**

Definition 3.3 (CNT-compressible sequence)**.** An infinite sequence of natural numbers is called a CNT-compressible sequence if it can be generated pointwise and exactly by a single static finite expression.

The intuition of compressibility is that the information of an entire infinite sequence is compressed into a finite static expression: the expression has finite length yet suffices to determine every position of the sequence.

3.4. **CNT-Incompressible Sequence.**

Definition 3.4 (CNT-incompressible sequence)**.** An infinite sequence of natural numbers is called a CNT-incompressible sequence if there exists no elementary or advanced static finite expression that generates the entire infinite sequence pointwise, without testing and without enumeration.

Incompressibility means that the information of the sequence cannot be concentrated into any finite static expression; the value at each position carries new information that cannot be predetermined by a unified formula. It must be stressed that even if an algorithmic program can generate the sequence step by step, this does not change its CNT-incompressibility, because an algorithm is dynamic rather than static generation.

3.5. **Cross-Domain Matching Proposition.**

Definition 3.5 (Cross-domain matching proposition)**.** A proposition P is called a cross-domain matching proposition if and only if it simultaneously satisfies the following three conditions:

(1) the proposition depends, in its subject matter, on a CNT-incompressible infinite sequence;
(2) its universal quantifier covers all the target infinite objects;
(3) there exists no unified static formula constructing each matching pair of elements.

The three conditions mean, respectively: first, the truth of the proposition is determined by the properties of an incompressible sequence; second, the proposition asserts something about every object in the target set; third, to verify the proposition for every object, one cannot use a unified static formula to complete all matchings at once, but must check object by object. Taken together, these conditions make the truth of the proposition depend on infinitely many mutually independent pieces of information, which structurally precludes a finite proof.

4. The Postulate System

Postulates are basic propositions accepted without proof, from which all subsequent conclusions are derived in one direction.

Postulate 4.1 (Finiteness Postulate)**.**

$$\forall P \ \big(\mathrm{Prov}(P) \Rightarrow \Box_{\mathrm{fin}} P\big).$$

The Finiteness Postulate asserts that every provable proposition admits a finite proof. It anchors the notion of proof in finiteness and excludes alleged proofs that depend on infinite reasoning.

Postulate 4.2 (Sequence Dichotomy Postulate)**.**

$$\forall S, \ \mathrm{Compressible}(S) \oplus \mathrm{Incompressible}(S).$$

The Sequence Dichotomy Postulate asserts that every infinite sequence of natural numbers is either a CNT-compressible sequence or a CNT-incompressible sequence, and not both.

Postulate 4.3 (Matching Support Core Postulate)**.**

$$\mathrm{Depends}(P, S) \wedge \mathrm{Incompressible}(S) \wedge \mathrm{CrossMatching}(P) \Rightarrow \neg\Box_{\mathrm{fin}}(P).$$

The Matching Support Core Postulate asserts that if a proposition P is a cross-domain matching proposition resting on an incompressible sequence S, then P admits no finite proof. Here $\mathrm{Depends}(P, S)$ denotes that the subject matter of P depends on the sequence S, i.e. the truth of P is determined by the properties of S. This postulate is the core of the CNT framework.

Postulate 4.4 (Prime Incompressibility Postulate)**.** The prime sequence is a CNT-incompressible sequence.

The Prime Incompressibility Postulate is the starting point of the framework. It asserts that the prime sequence cannot be generated as a whole by any static finite expression, and hence belongs to the class of incompressible sequences. This postulate is accepted without proof, as a first principle.

5. Basic Theorems

Theorem 5.1 (Non-existence of an Ideal Expression)**.** There exists no static ideal expression (elementary or advanced), free of primality testing, that can decide primality or generate the prime sequence.

Proof. Suppose that such a static ideal expression exists, which can decide primality or generate the prime sequence. Then, by the definition of a compressible sequence, any sequence generated pointwise and exactly by a single static finite expression is a CNT-compressible sequence; hence the prime sequence would be a CNT-compressible sequence. This directly contradicts the Prime Incompressibility Postulate (Postulate 4.4). Therefore the assumption is false, and no such ideal expression exists. □

Remark 5.1. Theorem 5.1 and Postulate 4.4 together characterize the incompressible nature of prime distribution and provide a premise for the subsequent characterization of Goldbach's conjecture. The "static ideal expression" is restricted to be free of primality testing and enumeration, a convention peculiar to the CNT framework.

6. Proposition Classification: Plausible-but-Unprovable Propositions

Definition 6.1 (Plausible-but-unprovable proposition)**.** A proposition P is called a plausible-but-unprovable proposition if and only if it simultaneously satisfies the following two conditions:

(1) it has no counterexample over its finitely known domain;

(2) because its information structure is beyond the finite limit, there exists no finite proof.

The first condition asserts that the proposition admits no counterexample within its finitely known domain (the finite range examined so far), and that its structural features and statistics point to truth, so that the proposition is highly *plausible* in experience. The second condition points out that the information structure on which it rests exceeds the carrying capacity of finite reasoning, so that no finite proof exists. Together the two conditions describe a special class of propositions: *highly plausible in experience, yet unprovable by finite mathematics.*

Remark 6.1 (Distinction from Gödelian undecidability)**.** In the Gödelian setting, an undecidable proposition is one that a system can neither prove nor refute, so that the system cannot judge its truth value. A CNT plausible-but-unprovable proposition, by contrast, has no counterexample over its finitely known domain and its structure points to truth; the issue is that this *plausibility* cannot be borne by finite logic to become a proof. Hence it is not a pending proposition but one that is highly plausible in experience yet unattainable as a finite proof owing to the transfinite character of its information structure.

7. The Characterization of Goldbach's Conjecture

7.1. **Formalization.**

Proposition 7.1 (Formalization)**.** Goldbach's conjecture admits the formal statement

$$\forall N \geq 2,\ \exists p, q \in \mathbb{P},\ p + q = 2N,$$

that is, for every integer $N \geq 2$, the even number $2N$ can be expressed as the sum of two primes.

7.2. **Classification Determination.**

Theorem 7.1 (Classification Determination)**.** Goldbach's conjecture is a standard CNT cross-domain matching proposition.

Proof. We verify the three conditions of the definition of a cross-domain matching proposition one by one. First, the proposition rests, in its subject matter, on the prime sequence, which by the Prime Incompressibility Postulate (Postulate 4.4) is a CNT-incompressible sequence; hence condition (i) holds. Second, the universal quantifier of the proposition covers all even numbers, i.e. it ranges over an infinite domain; hence condition (ii) holds. Third, by the Non-existence of an Ideal Expression Theorem (Theorem 5.1), there exists no static ideal expression free of primality testing that can generate the prime sequence or decide primality; hence there exists no unified static formula constructing a prime pair for every even number, and condition (iii) holds. All three conditions hold; by the definition, Goldbach's conjecture is a standard CNT cross-domain matching proposition. □

7.3. **Provability Derivation.**

Theorem 7.2 (Provability Derivation)**.**

$$\mathrm{Depends}(\mathrm{GC}, S_{\mathbb{P}}) \wedge \mathrm{Incompressible}(S_{\mathbb{P}}) \wedge \mathrm{CrossMatching}(\mathrm{GC}) \Rightarrow \neg\Box_{\mathrm{fin}}(\mathrm{GC}).$$

Proof. By the Classification Determination Theorem (Theorem 7.1), Goldbach's conjecture is a cross-domain matching proposition, and the prime sequence on which it rests is incompressible. Setting S to the prime sequence $S_{\mathbb{P}}$ and P to Goldbach's conjecture GC in the Matching Support Core Postulate (Postulate 4.3), the three premises $\mathrm{Depends}(\mathrm{GC}, S_{\mathbb{P}})$, $\mathrm{Incompressible}(S_{\mathbb{P}})$ and

CrossMatching(GC) are all satisfied; hence, by the Matching Support Core Postulate, Goldbach's conjecture admits no finite proof. □

Corollary 7.1. If Goldbach's conjecture has no counterexample over its whole domain, then it is a plausible-but-unprovable proposition in the CNT framework and admits no finite mathematical proof.

Proof. Suppose Goldbach's conjecture has no counterexample over its whole domain. By the Provability Derivation Theorem (Theorem 7.2), it admits no finite proof. Together with the fact that it has no counterexample over its finitely known domain, both conditions of the definition of a plausible-but-unprovable proposition (Definition 6.1) are satisfied; hence it is a plausible-but-unprovable proposition. □

Proposition 7.2. If Goldbach's conjecture has a counterexample, then it can be finitely refuted and does not belong to the class of plausible-but-unprovable propositions.

Proof. Suppose Goldbach's conjecture has a counterexample, i.e. there exists an even number that cannot be expressed as the sum of two primes. This counterexample is a finitely verifiable object, and verifying the non-existence of its prime-pair representation constitutes a finite proof. Hence the conjecture can be finitely refuted, so it does not satisfy the condition of admitting no finite proof, and it does not belong to that class. □

In summary, the provability status of Goldbach's conjecture can be characterized as follows: if it has no counterexample over its whole domain, then it admits no finite proof; if a counterexample is found, then it can be finitely refuted. The CNT framework does not decide its truth value but only characterizes its provability structure.

8. The Ultimate Meta-Conjecture of CNT

We fix the following notation. Let A denote the proposition Incompressible($S_{\mathbb{P}}$) (the prime sequence is CNT-incompressible, Postulate 4.4); let B denote the formula of the Matching Support Core Postulate (Postulate 4.3); let G denote Goldbach's conjecture GC.

Conjecture 8.1 (The Ultimate Meta-Conjecture of CNT)**.** In the standard model of the natural numbers,

$$(A \wedge B) \to \Big[\forall \mathcal{T},\ (\mathcal{T} \vdash G) \Rightarrow (\mathcal{T} \vdash A \wedge B)\Big].$$

The meaning of this conjecture is that, provided the incompressibility of the prime sequence and the Matching Support Core Postulate are true, for any axiom system $\mathcal{T}$, if $\mathcal{T}$ can finitely prove Goldbach's conjecture, then $\mathcal{T}$ necessarily logically entails all the core postulates of CNT, i.e. entails both A and B.

This assertion rules out the possibility of proving Goldbach's conjecture by bypassing the underlying logic of CNT. It maintains that any seemingly new axiom system, as long as it can prove Goldbach's conjecture, has its underlying logic equivalent to the core postulates of CNT; there is no proof path that circumvents the CNT restriction by replacing or adding axioms. Accordingly, the unprovability of Goldbach's conjecture is not a limitation of the framework but an intrinsic property of the information structure of the proposition itself.

Remark 8.1. The Ultimate Meta-Conjecture is a meta-mathematical conjecture whose status differs from that of Goldbach's conjecture itself: it quantifies over axiom systems and examines implication relations between provability claims. This conjecture has not been proved; any conclusion relying on it should be treated as conditional.

9. General Conclusions

Theorem 9.1 (General Theorem)**.** All domain-pairing, covering, and matching propositions of incompressible infinite sequences admit no finite mathematical proof.

Proof. By the Matching Support Core Postulate (Postulate 4.3), every cross-domain matching proposition resting on an incompressible sequence admits no finite proof. The General Theorem is a direct consequence of this postulate. □

Proposition 9.1 (Conditional conclusion, dependent on the Ultimate Meta-Conjecture)**.** If the Ultimate Meta-Conjecture (Conjecture 8.1) holds, then such propositions also cannot bypass the CNT postulate system, i.e. any axiom system that can finitely prove them necessarily entails the core postulates of CNT.

Explanation. This conditional conclusion presupposes the truth of the Ultimate Meta-Conjecture. Since the latter has not been proved so far, this conclusion is conditional rather than established; until the Ultimate Meta-Conjecture is proved, it must not be regarded as an established mathematical conclusion. □

10. Conclusion

The CNT framework, with the dichotomy of information-compression properties of sequences at its core, establishes a deductive chain from sequence structure to proposition provability. The framework does not decide the truth value of any number-theoretic proposition but only characterizes its finite provability; every cross-domain matching proposition resting on an incompressible sequence admits no finite proof in the CNT framework. Goldbach's conjecture, as a typical representative of such propositions, has its provability status characterized as follows: if it has no counterexample over its whole domain, it admits no finite proof; if a counterexample is found, it can be finitely refuted. This framework reduces the problem of provability to the problem of the information structure of sequences, thereby providing a unified theoretical basis for understanding the boundary of proofs of infinite number-theoretic propositions.